

ASTALAIR 10 mg Tablet**LEUKOTRIENE RECEPTOR ANTAGONIST**

Each film coated tablet contains:
Montelukast sodium eq. to Montelukast 10 mg

❖ **PRODUCT DESCRIPTION:**

Light peach, square-shaped, biconvex, film coated tablet with engraved M10 on one side and plain on the other

❖ **MECHANISM OF ACTION:****Pharmacology**

The cysteinyl leukotrienes (LTC₄, LTD₄, LTE₄) are potent inflammatory eicosanoids released from various cells including mast cells and eosinophils. These important pro-asthmatic mediators bind to cysteinyl leukotriene (CysLT) receptors. The CysLT type-1 (CysLT₁) receptor is found in the human airway (including airway smooth muscle cells and airway macrophages) and on other pro-inflammatory cells (including eosinophils and certain myeloid stem cells). CysLTs have been correlated with the pathophysiology of asthma and allergic rhinitis. In asthma, leukotriene-mediated effects include bronchoconstriction, mucous secretion, vascular permeability, and eosinophil recruitment. In allergic rhinitis, CysLTs are released from the nasal mucosa after allergen exposure during both early- and late-phase reactions and are associated with symptoms of allergic rhinitis. Intranasal challenge with CysLTs has been shown to increase nasal airway resistance and symptoms of nasal obstruction.

Montelukast is an orally active compound which binds with high affinity and selectivity to the CysLT₁ receptor.

Pharmacokinetics**Absorption**

Montelukast is rapidly absorbed following oral administration. For the 10-mg film coated tablet, the mean peak plasma concentration (C_{max}) is achieved 3 hours (T_{max}) after administration in adults in the fasted state. The mean oral bioavailability is 64%. The oral bioavailability and C_{max} are not influenced by a standard meal.

Distribution

Montelukast is more than 99% bound to plasma proteins. The steady-state volume of distribution of Montelukast averages 8-11 liters.

Biotransformation

Montelukast is extensively metabolized.

The contribution of metabolites to the therapeutic effect of Montelukast is minimal.

Elimination

The plasma clearance of Montelukast averages 45 mL/min in healthy adults. Following an oral dose of radiolabeled Montelukast, 86% of the radioactivity was recovered in 5-day fecal collections and < 0.2% was recovered in urine. Coupled with estimates of Montelukast oral bioavailability, this indicates that Montelukast and its metabolites are excreted almost exclusively via the bile.

Characteristics in Patients

No dosage adjustment is necessary for the elderly or mild to moderate hepatic insufficiency. Studies in patients with renal impairment have not been undertaken. Because Montelukast and its metabolites are eliminated by the biliary route, no dose adjustment is anticipated to be necessary in patients with renal impairment. There are no data on the pharmacokinetics of Montelukast in patients with severe hepatic insufficiency (Child-Pugh score > 9).

With high doses of Montelukast (20- and 60-fold the recommended adult dose), decrease in plasma Theophylline concentration was observed. This effect was not seen at the recommended dose of 10 mg once daily.

❖ **INDICATION¹:**

- For the prophylaxis and chronic treatment of asthma in adults and adolescents 15 years of age and older.
- ASTALAIR 10 is indicated in adults and adolescents 15 years of age and older for the relief of daytime and night time symptoms of seasonal allergic rhinitis.

❖ **DOSAGE AND ADMINISTRATION¹:**

ASTALAIR 10 should be taken once daily.

For asthma, the dose should be taken in the evening. For allergic rhinitis, the time of administration may be individualized to suit patient needs. Patients with both asthma and allergic rhinitis should take only one tablet daily in the evening.

Adults 15 Years of Age and Older with Asthma and/or Seasonal Allergic Rhinitis

The dosage for adults 15 years of age and older is one 10-mg tablet daily.

General Recommendations

The therapeutic effect of ASTALAIR 10 on parameters of asthma control occurs within one day. ASTALAIR 10 Tablet can be taken with or without food. Patients should be advised to continue taking ASTALAIR 10 while their asthma is controlled, as well as during periods of worsening asthma. No dosage adjustment is necessary for the elderly, for patients with renal insufficiency, or mild-to-moderate hepatic impairment, or for patients of either gender.

Montelukast is a long term-controller medication which should not be substituted for short acting beta-agonists. It is effective alone or in combination with other prophylactic agent. Montelukast is a preventive agent, which should be used in addition to other drugs for the management of asthma.

Therapy with ASTALAIR 10 in Relation to Other Treatments for Asthma

ASTALAIR 10 can be added to a patient's existing treatment regimen.

Reduction in Concomitant Therapy

Bronchodilator treatments: ASTALAIR 10 can be added to the treatment regimen of patients who are not adequately controlled on bronchodilator alone. When a clinical response is evident (usually after the first dose), the patient's bronchodilator therapy can be reduced as tolerated.

Inhaled corticosteroids: Treatment with ASTALAIR 10 provides additional clinical benefit to patients treated with inhaled corticosteroids. A reduction in the corticosteroid dose can be made as tolerated. The dose should be reduced gradually with medical supervision. In some patients, the dose of inhaled corticosteroids can be tapered off completely. ASTALAIR 10 should not be abruptly substituted for inhaled corticosteroids.

❖ **WARNING AND PRECAUTION:**

Patients should be advised never to use oral Montelukast to treat acute asthma attacks and to keep their usual appropriate rescue medication for this purpose readily available. If an acute attack occurs, a short-acting inhaled beta-agonist should be used. Patients should seek their doctors' advice as soon as possible if they need more inhalations of short-acting beta-agonists than usual.

Montelukast should not be substituted for inhaled or oral corticosteroids.

There are no data demonstrating that oral corticosteroids can be reduced when Montelukast is given concomitantly.

In rare cases, patients on therapy with anti-asthma agents including Montelukast may present with systemic eosinophilia, sometimes presenting with clinical features of vasculitis consistent with Churg-Strauss syndrome, a condition which is often treated with systemic corticosteroid therapy. These cases usually, but not always, have been associated with the reduction or withdrawal of oral corticosteroid therapy. The possibility that leukotriene receptor antagonists may be associated with emergence of Churg-Strauss syndrome can neither be excluded nor established. Physicians should be alert to eosinophilia, vasculitic rash, worsening pulmonary symptoms, cardiac complications, and/or neuropathy presenting in their patients. Patients who develop these symptoms should be reassessed and their treatment regimens evaluated.

Treatment with Montelukast does not alter the need for patients with Aspirin-sensitive asthma to avoid taking Aspirin and other non-steroidal anti-inflammatory drugs.

This medicinal product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Warning²:

This preparation contains Sodium metabisulfite that may cause serious allergic-type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulfites.

❖ **PREGNANCY AND LACTATION:****Pregnancy**

Limited data from available pregnancy databases do not suggest a causal relationship between Montelukast and malformations (i.e. limb defects) that have been rarely reported in worldwide post marketing experience.

Montelukast may be used during pregnancy only if it is considered to be clearly essential.

PM Specification 	Product	Code No.	Dimension	Packaging Type	Thickness
	ASTALAIR 10	IAMY 0061	W. 19 x L 24.5 cm.	Wood Free Paper กระดาษปอนด์	60 g (0.08 mm.)
Designed by: (PDM, ASD, PDCM, PDC-A, PDC-M, PDC-A) Checked by: (PDM/ASPD) Approved by: (MD (Only Pattern and Color))					
Approved by: PLC: (Date) LRA: (Date) IRA: (Date) QCC-PM: (Date) CUSTOMER/SALES DEPT: (Date)					

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Lactation

It is not known if Montelukast is excreted in human milk.

Montelukast may be used in nursing mothers only if it is considered to be clearly essential.

✧ **SIDE EFFECT:**

Tabulated list of Adverse Reactions

Adverse reactions are listed by System Organ Class and specific Adverse Reactions, in the table below:

System Organ Class	Adverse Reactions	Frequency Category
Infections and infestations	Upper respiratory infection	Very Common
Blood and lymphatic system disorders	Increased bleeding tendency	Rare
	Thrombocytopenia	Very rare
Immune system disorders	Hypersensitivity reactions including anaphylaxis	Uncommon
	Hepatic eosinophilic infiltration	Very rare
Psychiatric disorders	Dream abnormalities including nightmares, insomnia, somnambulism, anxiety, agitation including aggressive behavior or hostility, depression, psychomotor hyperactivity (including irritability, restlessness, tremor)	Uncommon
	Disturbance in attention, memory impairment, tic	Rare
	Hallucinations, disorientation, suicidal thinking and behavior (suicidality)	Very rare
Nervous system disorders	Dizziness, drowsiness, paraesthesia/hypoesthesia, seizure	Uncommon
Cardiac disorders	Palpitations	Rare
Respiratory and thoracic and mediastinal disorders	Epistaxis	Uncommon
	Churg-Strauss Syndrome (CSS)	Very rare
	Pulmonary eosinophilia	Very rare
Gastro-intestinal disorders	Diarrhea, nausea, vomiting	Common
	Dry mouth, dyspepsia	Uncommon
Hepatobiliary disorders	Elevated levels of serum transaminase (ALT, AST)	Common
	Hepatitis (including cholestatic, hepatocellular, and mixed-pattern liver injury)	Very rare
Skin and subcutaneous tissue disorders	Rash	Common
	Bruising, urticaria, pruritus	Uncommon
	Angioedema	Rare
	Erythema nodosum, erythema multiforme	Very rare
Musculoskeletal and Connective tissue disorders	Arthralgia, myalgia including muscle cramps	Uncommon
Renal and urinary disorders	Enuresis in children	Uncommon
General disorders and administration site conditions	Pyrexia	Common
	Asthenia/fatigue, malaise, edema	Uncommon

Post-marketing experience

¹Blood and lymphatic system disorders: Thrombocytopenia

²Psychiatric disorders: Obsessive-compulsive symptoms

✧ **CONTRAINDICATION:**

Hypersensitivity to the active substance or to any of the excipients.

✧ **DRUG INTERACTION:**

Montelukast may be administered with other therapies routinely used in the prophylaxis and chronic treatment of asthma.

The recommended clinical dose of Montelukast do not have clinically important effects on the pharmacokinetics of the following drugs: Theophylline, Prednisone, Prednisolone, oral contraceptives (Ethinyl estradiol/ Norethindrone 35/1), Terfenadine, Digoxin and Warfarin.

Since Montelukast is metabolized by CYP 3A4, caution should be exercised, particularly in children, when Montelukast is coadministered with inducers of CYP 3A4, such as Phenytoin, Phenobarbital and Rifampicin.

Montelukast is not anticipated to markedly alter the metabolism of drugs metabolized by this enzyme (e.g. Paclitaxel, Rosiglitazone, and Repaglinide).

✧ **OVERDOSE AND TREATMENT:**

No specific information is available on the treatment of overdose with Montelukast. In chronic asthma studies, Montelukast has been administered at doses up to 200 mg/day to patients for 22 weeks and in short term studies, up to 900 mg/day to patients for approximately one week without clinically important adverse experiences.

There have been reports of acute overdose in post-marketing experience and clinical studies with Montelukast. These include reports in adults and children with a dose as high as 1000 mg (approximately 61 mg/Kg in a 42 month old child). The clinical and laboratory findings observed were consistent with the safety profile in adults and pediatric patients. There were no adverse experiences in the majority of overdosage reports. The most frequently occurring adverse experiences were consistent with the safety profile of Montelukast and included abdominal pain, somnolence, thirst, headache, vomiting, and psychomotor hyperactivity.

It is not known whether Montelukast is dialyzable by peritoneal- or hemodialysis.

✧ **STORAGE:**

Store at temperature of not more than 30°C.

✧ **DOSAGE FORM AND PACKAGING AVAILABLE:**

10 mg Tablet, Blister 4x7's & 10x10's

✧ **DATE OF REVISION:**

May 30, 2022

Manufactured by:
UNISON LABORATORIES CO., LTD.
 39 Moo 4, Klong Udomcholjorn, Muang Chachoengsao,
 Chachoengsao 24000 Thailand

C-MY/300522-05 (AR)

(www.emc-medicines.org.uk; ¹DRGD_250719; ²DRGD_180522)

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